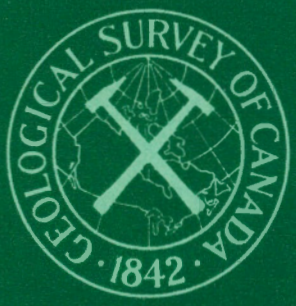
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GEOLOGICAL
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BULLETIN 84

PALAEOMAGNETISM OF PART OF THE PURCELL SYSTEM IN SOUTHWESTERN ALBERTA AND SOUTHEASTERN BRITISH COLUMBIA

R. F. Black

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1963

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By
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PREFACE

This bulletin deals with the application of palaeomagnetic studies to several late Precambrian formations of the Purcell System in the eastern Rocky Mountains and the interpretation of the results. The palaeomagnetic data reported here indicate that the position of the geomagnetic pole at the time these rocks were magnetized was vastly different from that of the present pole. By comparison with the geomagnetic pole positions reported from other late Precambrian formations in North America, it is concluded that the Purcell System may be of late Keweenawan age.

J. M. HARRISON,
Director, Geological Survey of Canada

OTTAWA, January 11, 1961

Bulletin 84—Paläomagnetismus in dem im Südwesten von Alberta und Südosten von Britisch-Kolumbien gelegenen Teile des Purcellsystems. Von R. F. Black

Dieses Bulletin behandelt die Anwendung paläomagnetischer Untersuchungen und die Auslegung der Ergebnisse der Untersuchung mehrerer spätpräkambrischer Formationen, die zu dem Purcellsystem im östlichen Teil des Felsengebirges gehören.

Бюллетень 84 — Палеомагнетизм Пурселлской системы в югозападной Альберте и юговосточной Британской Колумбии. Автор: Р. Ф. Блэк.

Этот бюллетень посвящен применению палеомагнетических исследований и интерпретации результатов, полученных при исследовании нескольких докембрийских свит, составляющих часть Пурселлской системы в восточных Скалистых Горах.

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PALAEOMAGNETISM OF PART OF THE PURCELL SYSTEM IN SOUTHWESTERN ALBERTA AND SOUTHEASTERN BRITISH COLUMBIA

Abstract

The Precambrian Purcell System (Belt) is separated from overlying Cambrian strata by a major unconformity, which leaves the age of the Purcell in doubt. Absolute age determinations on associated material range from 710 to 1,200 million years, and are at best a minimum range.

An analysis of oriented samples from six Purcell formations was made in the hope that an age could be assigned from the magnetic data obtained. Final results indicated magnetic stability in four of the sedimentary formations. Calculations based on a study of seventy-five samples collected from the stable formations revealed no significant difference in the mean direction of the magnetic moments between the lowest (Waterton) and the uppermost (Kintla) stable formations. An estimate of the coordinates of the ancient geomagnetic pole for Purcell time is calculated to be 148.0°W longitude and 10.0°S latitude.

Estimates of the age of the Purcell System are given by a comparison of its pole position with those for other North American late Precambrian formations. DuBois (1962) suggests an early Keweenawan age based on polar wandering interpretation that the Belt Series is definitely older than the Duluth Gabbro of middle Keweenawan age (1,120 m.y.). The author suggests a possible upper Keweenawan age, based on polar wandering interpretation that places the age of the Purcell at approximately 900 to 1,000 million years—younger than the Duluth Gabbro.

Résumé

Le système Purcell d'âge précambrien est séparé des strates cambriennes surjacentes par une importante discordance de stratification, ce qui rend incertain l'âge du système Purcell. Des déterminations d'âge absolu faites sur des échantillons provenant du système Purcell ont donné des résultats variant entre 710 et 1,200 millions d'années, et c'est là tout au plus une variation minimum.

On a étudié l'aimantation d'échantillons orientés en provenance des six formations du système Purcell dans l'espoir d'en déterminer l'âge à l'aide des données ainsi obtenues. Les résultats définitifs ont indiqué que quatre des formations sédimentaires étaient magnétiquement stables. Les calculs fondés sur une étude de soixante-quinze échantillons prélevés au sein des formations stables n'ont révélé aucune différence appréciable dans la direction moyenne des moments magnétiques entre la formation la moins stable (Waterton) et la formation la plus stable (Kintla). Les coordonnées de l'ancien pôle géomagnétique au temps du Purcell ont été évaluées par calculs comme étant: longitude 148.0°W, latitude 10.0°S.

On a évalué l'âge du système Purcell en comparant la position du pôle de cette période aux positions du pôle des autres formations nord-américaines

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de la fin du Précambrien. DuBois (1962) opine en faveur d'un âge se situant au début du Keweenawien en se basant sur la théorie de la dérive des pôles; il soutient que la série Belt est nettement plus ancienne que le gabbro de Duluth, qui remonte au Keweenawien moyen (1,120 m.a.). L'auteur du présent Bulletin croit possible que le Purcell remonte au Keweenawien supérieur, compte tenu de l'interprétation de la dérive polaire, suivant laquelle l'âge du Purcell se situerait entre 900 et 1,000 millions d'années, ce qui signifierait que le système est plus récent que le gabbro de Duluth.

INTRODUCTION

This investigation was an attempt to date Purcell[1] rocks by a comparison of palaeomagnetic results for a number of Purcell formations with those for other late Precambrian formations of North America whose ages are relatively well known. The length of time between the deposition of rocks of Purcell age and those of Cambrian age cannot be resolved because of a lack of absolute dating of the rocks involved.

The presence of primitive life and the low grade of metamorphism in the Belt Series has led some observers to believe that the Beltian rocks are not a great deal older than the overlying Cambrian rocks. However, a more detailed study has shown that a major unconformity exists between the two successions (*see* p. 2), and this has led to considerable discussion and doubt as to the age of the Belt Series.

The field work for this palaeomagnetic investigation was done in the summer of 1959, in the extreme eastern part of the Rocky Mountains of southeastern British Columbia and southwestern Alberta; specifically the Lewis and Clark Ranges between latitudes 49°00′ and 49°30′, and longitudes 114°00′ and 114°45′.

Acknowledgments

This investigation was commenced by P. M. DuBois, formerly on the staff of the Geological Survey of Canada, and was taken over during the field season by the writer who carried out most of the collection and the laboratory work. S. R. Sopher ably assisted in the field.

General Geology of Belt Series

Parts of the Rocky Mountains of Montana, Idaho, British Columbia, and southwestern Alberta are composed of a large number of conformable sedimentary formations that have been termed the Belt Series by Walcott (1899)[2] after the Little Belt Mountains of Montana. Fossil evidence pointing to the exact age of these rocks is very scarce, but it was noted that life during the time of deposition was of a rudimentary nature. Parts of the limestone strata are formed by extensive deposits of calcareous algae (Fenton and Fenton, 1937) occurring in individual masses at various stratigraphic levels. Evidence of worm tubes and trails of crawling invertebrate in these rocks is rare. Generally the exposed sections reveal a total thickness of from 12,000 to 25,000 feet of Beltian rocks but in places, as in the Cabinet Range of extreme western Montana, depositions are as much as

[1]Although the terms Belt and Purcell are not synonymous in every respect, they are interchangeable as used in this paper. Inasmuch as the description of the general geology includes observations made by American workers the term Belt has been used, but where specific reference is made pertaining to Canadian geology the term Purcell is used.

[2]Names and/or dates in brackets are those of references cited in the Bibliography.

35,000 feet thick. In southwestern Alberta, where this investigation was carried out, a total thickness of approximately 8,000 feet from the lowest part of the Waterton Formation to the top of the Kintla Formation has been measured.

The unmetamorphosed Beltian strata are separated by a major unconformity from a basement complex of ancient crystalline rocks that are partly metamorphic and partly of plutonic origin (Schuchert and Dunbar, 1941). The Belt Series is unconformably overlain by the Flathead Formation of possible Middle Cambrian age and this in turn by dated trilobite-bearing middle Cambrian strata, the Wolsey Formation (Deiss, 1935). It has been proposed, from regional mapping of the unconformity, that after uplift occurred as much as 20,000 feet of Beltian rocks were removed during Lipalian time. Only gentle warping of Beltian strata occurred during uplift, and as a result the contact between Cambrian and Beltian strata in many local outcrops appears parallel. Regionally, however, it is apparent that a great thickness of Beltian rocks was removed and thus a major hiatus occurred between the deposition of the Belt Series and that of the Cambrian rocks.

A final deformation of the miogeosyncline during very late Cretaceous and early Tertiary time, termed the Laramide Revolution, elevated the strata thousands of feet with attendant large-scale thrust faulting and folding.

Geology of Area Investigated

The Precambrian rocks found in the area (*see* Fig. 1) consist of variously coloured, indurated arenaceous, argillaceous, and calcareous strata in which minor sills and lava flows are intercalated. The occurrence of crossbedding, ripple-marks, mud-cracks, and the well oxidized condition of some of the beds are indicators that the environment of deposition was shallow to very shallow. The clastic material forming these beds was probably derived from a crystalline basement complex east and northeast of the area (Reesor, 1957). A large section of the Purcell System has undergone an eastward to northeastward displacement, a minimum distance of 25 miles (Price, 1959) in the Lewis Thrust Plate relative to the foot-wall strata. In the area the Purcell rocks are in thrust fault contact with Cretaceous strata. Structural evidence suggests that this large thrust sheet has moved relatively eastward as a unit and has been elevated thousands of feet.

Geological maps by Price (1959), Norris (1959), Hage (1943), and Douglas (1952) were used to find sampling areas within the various formations. Exposed thicknesses of Purcell strata are given in Table I.

Table I

Formation Thickness (in feet) of the Purcell System Exposed in Southeastern British Columbia and Southwestern Alberta

Map-Area	Waterton (Douglas, 1952)	Beaver Mines (Hage, 1943)	Carbondale (Norris, 1959)	Flathead (Price, 1959)
Kintla C	Top not	1,000	460	400
Kintla B	exposed	500	510	500
Kintla A	1,500	800	820	650
Sheppard	600	470	440	120–160
Purcell Lava	200	350	295	320
Siyeh	3,000	3,000	1,650	1,130
Grinnell	750	760	630	350
Appekunny	1,100	1,600	2,100	1,700
		Base not exposed		
Altyn	500		480	1,250
Waterton	617		440	1,500
	Base not exposed		Base not exposed	Base not exposed
Total	8,267	8,480	7,825	7,960

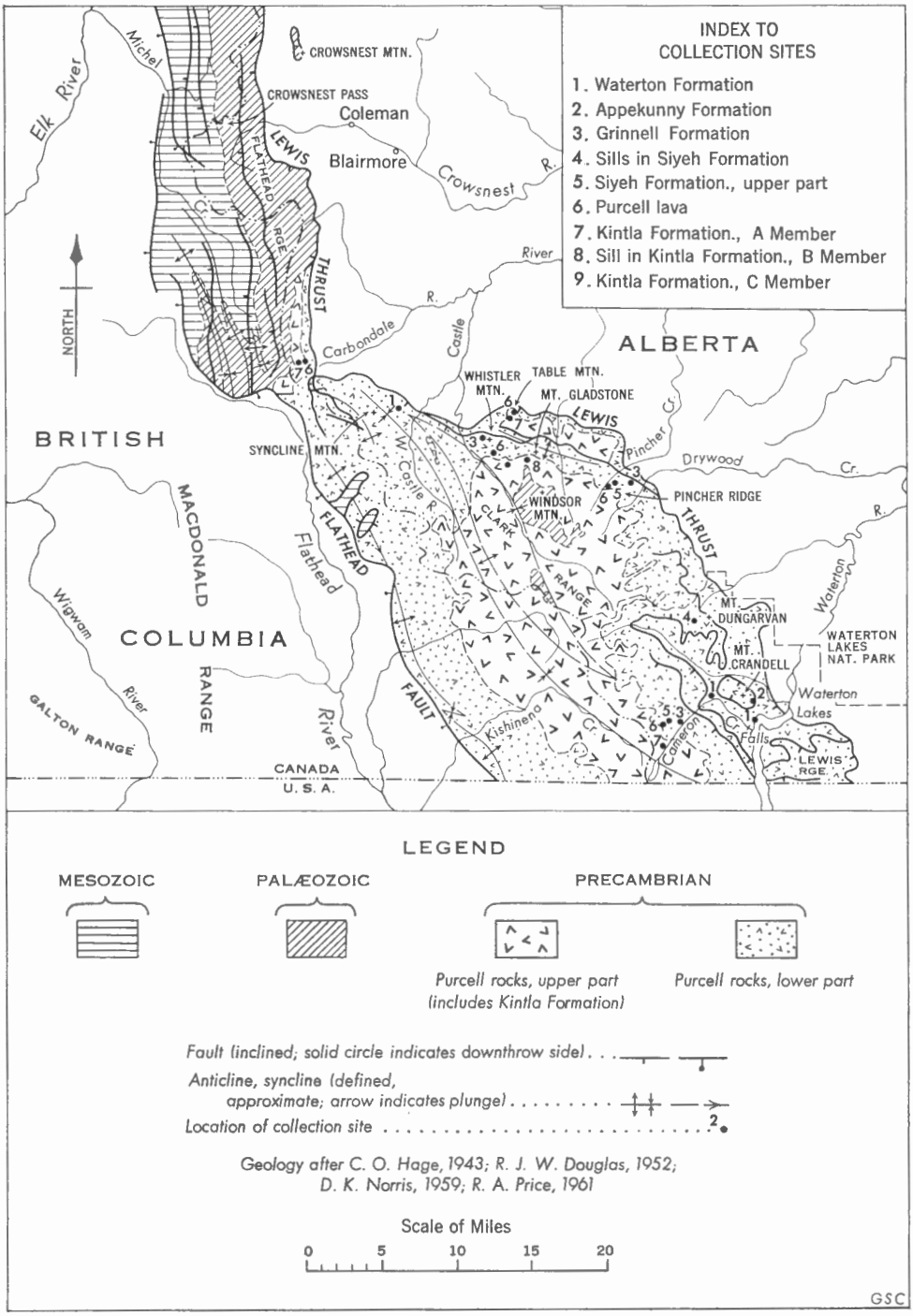


Figure 1. ***Generalized geological map of part of the Lewis Thrust Plate showing collection sites from the Purcell System*** **(after Norris and Black, 1962).**

TECHNIQUES

Collection of Samples[1]

Samples of fresh, unweathered, fine-grained, red to pink sandstones, siltstones, and dolomites, as well as fine- to medium-grained crystalline igneous rocks were collected in pairs a few feet apart. This was not always possible because of such features as friability of the argillites, thinness of bedding, or irregular surfaces. The sample was oriented by marking a horizontal line on a bedding plane or smooth joint face with a china marking pencil and hand level. An arrow head and dip symbol were appropriately marked on the horizontal line and this surface designated the reference plane of the oriented sample. The orientation marks were preserved by covering the sample with a coat of clear plastic Krylon spray. For sedimentary rocks, a Brunton compass was used to determine the azimuth of the reference arrow in degrees east of magnetic north, and the clinometer to determine the angle of dip from the horizontal to the oriented surface. Where the orientation marks had been placed on a joint face, a measurement of the strike and dip of bedding in the vicinity was taken so that corrections of the magnetic moment to the original bedding could be computed. For igneous rocks a solar compass adjusted to the appropriate sun time and latitude was used to measure the azimuth of the horizontal arrow on the oriented surface. This eliminated the introduction of error due to possible deflection of the compass needle by the ferromagnetic minerals within the outcrop.

Preparation and Measurement of Samples

The techniques followed in the preparation and magnetization measurements of oriented rock samples are fully described elsewhere in the literature, and, therefore, only briefly mentioned here.

The field sample was sliced with a diamond blade into a 1-inch slab, from which two 1-inch cubes were cut. One of the cube faces is necessarily in a plane parallel with the original collection plane and its edges are parallel with the reference arrow. A small reference arrow appropriately oriented is marked on this surface of the cube. The magnetization of the sedimentary rocks and of some of the more weakly magnetized igneous rocks was determined with a sensitive astatic magnetometer put at the disposal of the writer by the Dominion Observatory (DuBois, 1962). The igneous rocks that bear a more intense magnetic moment were processed in a spinner-type or rock generator magnetometer (Larochelle, 1962).

---

[1]A sample refers to an oriented piece of rock from which two specimens in the form of cubes are cut. "Locality" refers to a general area, whereas "collecting site" refers to a particular place of collection.

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From the measurements of two cubes a mean declination and inclination of the remanent magnetic moment was computed for each sample.

Isolation of Significant Components of Magnetization

It has been observed by Thellier and Rimbert (1954) that the measured remanent magnetic moment in a rock sample may not represent the true direction of magnetization acquired by the rock at the time of its formation, due to secondary components that may have been later introduced into the rock. The most common of these secondary components, the isothermal, is introduced at low temperatures by the action of the earth's changing magnetic field. Other components may be the result of chemical alteration, of reheating of the minerals contained in the rock, or of the action of lightning. Therefore, any rock for which a magnetic moment is to be determined may have for its magnetization a combination of its original magnetization and one or more of the secondary components. The magnetization vector measured is the vector sum of these components and the result could be quite different from the original magnetization.

The removal of the unstable isothermal component can sometimes be achieved by subjecting the specimen to the influence of an alternating magnetic field which is slowly reduced to zero in a magnetic field free space. This is to prevent the formation of a stable anhysteritic component (Thellier, 1946) in the specimen. A magnetic field free space is most conveniently obtained at the centre of a system of Helmholtz coils through which an appropriate current is allowed to circulate in order to cancel the earth's magnetic field. Also, to prevent the introduction of random magnetic vectors into the specimen it is essential that the amplitude of the alternating field be removed slowly and smoothly from the specimen. This is readily achieved by raising the alternating current coil slowly away from the specimen. The instrument used for this study has been described in detail by Larochelle (1962). The procedure of alternating current demagnetization is hereafter called magnetic washing.

To eliminate an excessive amount of time-consuming measurements, one cube from each sample was magnetically washed and its direction of magnetization was again determined. If the direction of the magnetic vector left in the specimen deviated from the original direction by more than 5 degrees of arc on a great circle then the second cube was washed in the same way. The specimens were submitted to alternating fields of increasing maximum amplitude until no significant change took place in the magnetization direction or until a maximum peak field of 312 oersteds had been applied.

Chemical alteration of ferromagnetic minerals and/or reheating of a rock would produce remanent magnetization indicative of the earth's field attitude at the time of metamorphism. However, if the alteration is slow or incomplete, the remanent moments for a group of samples from a formation will be randomly oriented, as the earth's magnetic field changes continuously. It is evident that if these components as well as an isothermal component are superimposed in a rock over its original magnetization a very complex magnetization will result and the

probability of measuring the desired original component becomes increasingly less. For this reason all samples were subjected to the influence of an alternating field to remove any isothermal components. A study of representative samples of Purcell Lava in thin section revealed that alteration of the ferromagnetic minerals had taken place. Consequently, there is the possibility that a chemical magnetization exists in the specimens.

Most of the sedimentary rocks were not significantly affected by the magnetic washing, which indicates that they contain no significant isothermal component. Some of the igneous rocks, however, showed large changes in the direction of magnetization as well as in their polarity after they had been magnetically washed. This is discussed on page 20.

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DESCRIPTION OF RESULTS

The directions of magnetization reported in this paper are defined by the letters D and I, D being the declination measured in degrees east of geographic north; I the inclination, which is positive in the downward direction and negative in the upward direction. Stereographic projections have been used for the graphical representation of the results. Full circles represent magnetic vectors pointing downward and open circles represent magnetic vectors pointing upward. The mean direction for a group of samples is represented by a slightly enlarged circle including a cross. A solid triangle plotted on the lower hemisphere represents the present theoretical magnetic dipole field direction at the collection locality.

The statistical parameters computed for the various groups of measurements are the usual Θ, K, and N, as defined by Fisher (1953); Θ is the circle of confidence calculated at the 95 per cent level of confidence; K and N represent a measure of dispersion or precision and the number of samples, respectively. The circles of confidence are represented graphically as broken line circles on the upper hemisphere and solid line circles on the lower hemisphere.

The mean declination and inclination for a group of samples collected from a formation reflect the attitude of the earth's magnetic field at the collection locality at the time the rocks were magnetized. If the mean direction of magnetization is determined for a sufficient thickness of rock, the elapsed length of time is sufficient to cancel secular variations, it then may be assumed that the magnetic field that produced this mean direction is that produced by an axial geocentric dipole. By use of the dipole formula[1] it is then possible to compute, from the mean direction of magnetization for a formation and its collecting site coordinates, the geographic coordinates or pole position of the geomagnetic pole produced by the assumed dipole. The quantities $d\psi$ and $d\chi$ are the semi-axes of the elliptical area of confidence round the pole at a probability of 95 per cent, $d\psi$ is the colatitude direction and $d\chi$ perpendicular to it. As seen in the discussion of the results, these errors are only given when a group of formations are summed together to ensure the elimination of secular variation.

All the measurements reported in this paper are with respect to the bedding plane, that is when the bedding surface was in an assumed horizontal position at the time of deposition. In igneous rocks, or sedimentary rocks not sampled on a bedding plane, corrections were made to refer the magnetization directions to the bedding of the formation in the near vicinity.

[1] cot. $p = \frac{1}{2} \tan I$, p is the angular distance along the great circle from the collecting site to the geomagnetic pole and I is the inclination of the field at the collection locality.

Waterton Formation

Twenty samples from Waterton strata were collected from two main exposures at widely separated localities (Fig. 1, No. 1). Seventeen were used in the final analysis, three having been rejected for various reasons as indicated at the bottom of Table II.

The Waterton strata collected are mainly red to pale reddish brown dolomite and limestone interbedded with pale grey and green dolomites and argillites and grey limestones. The Waterton Formation is the lowest formation of the Purcell System that is exposed in the investigated area.

Table II

Magnetic Measurements for Waterton Formation

Sample	Original		Final		Collection Site (1)
	D	I	D	I	
DGN 12B	215.5	+32.0	203.5	+26.0	Northwest side of West Castle River
[1] 12S	185.5	+28.0	199.5	+33.0	at foot of Syncline Mtn., approx-
13B	207.0	+57.5	188.0	+52.5	imately 4 miles southwest of junc-
13S	220.5	+40.0	200.5	+31.0	tion of west and south branches
[1] 14B	15.5	+ 6.5	216.5	+26.0	of Castle River.
14S	229.5	+16.5	211.0	+20.5	
15B	224.0	+21.0	213.0	+38.0	
15S	212.0	+38.0	183.5	+34.0	
[1] 16B	208.0	+17.0	226.5	+18.5	Lat. 49°21′30″
16S	213.0	+47.0	181.5	+52.5	Long. 114°24′30″
17B	207.0	+36.0	195.5	+34.0	
17S	215.5	+29.5	178.5	+30.0	
18B	208.0	+29.0	200.5	+29.0	
18S	180.0	+28.5	186.5	+29.0	
					3 miles from Cameron Falls on Cameron Creek road.
DGW 3B	217.5	+21.5	217.5	+21.5	Lat. 49°05′, long. 113°57′.
DGW 8P	208.0	+58.0	221.0	+51.5	Collected at the foot of Cameron
9P	245.5	+25.0	200.0	+23.5	Falls, Waterton Lakes National Park. Lat. 49°03′, long. 113°55′.
Mean	214.0	+36.0	202.0	+33.0	

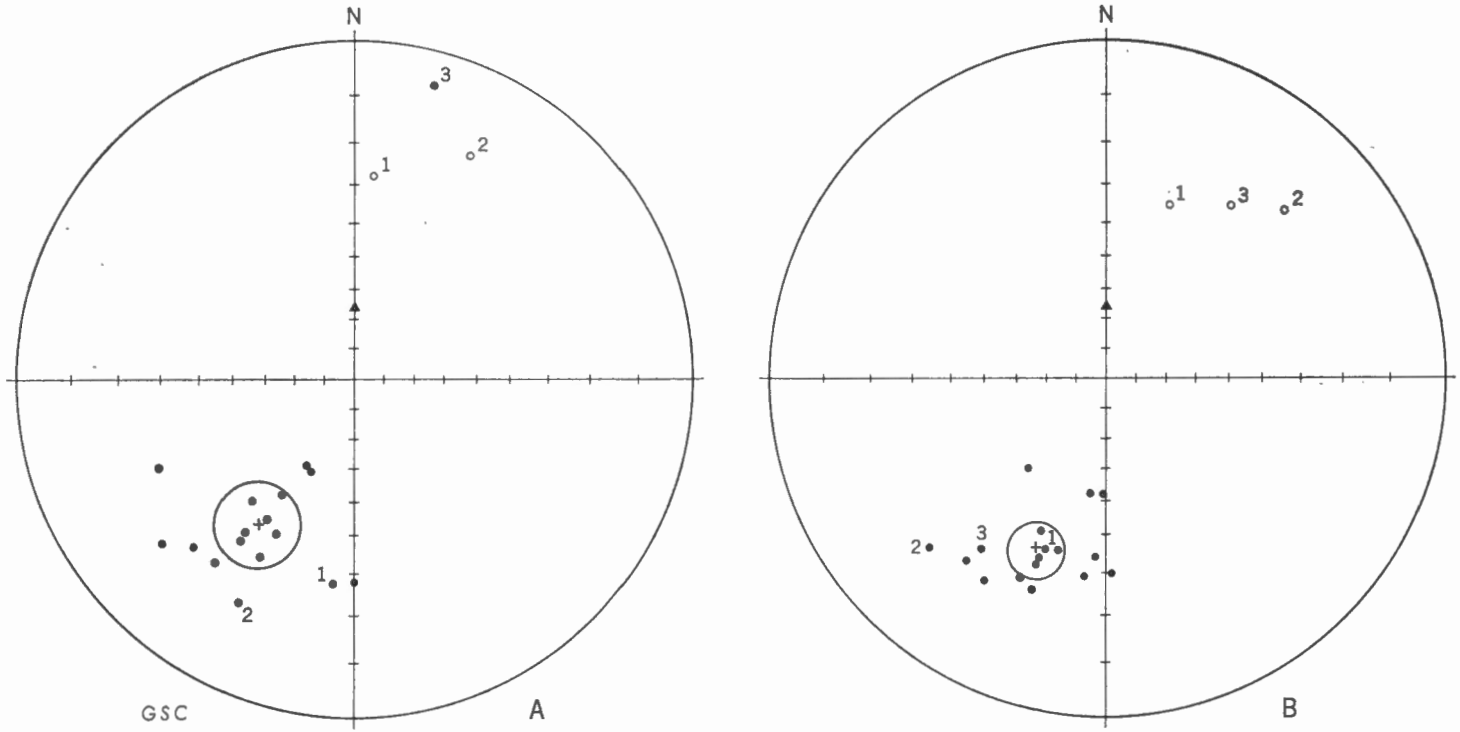
[1] Has been adjusted for statistical purposes as originally negative and 180° less in declination.

Rejections

DGW 3P—A pale green-grey sample, too weak to be measured accurately with the astatic magnetometer.

DGW 8B & 9B—Weakly magnetized, and from poor outcrops that were possibly out of place.

The remanent magnetization direction of the specimens was altered by magnetic washing, with a shift of 12°E in the mean declination after the specimens had been subjected to a field of 156 oersteds. The inclination was not significantly



LEGEND

North pole on lower hemisphere . . . . . ●

North pole on upper hemisphere . . . . . o

Mean direction with circle of confidence, north pole on lower hemisphere . . . . . +

Present theoretical dipole . . . . . ▲

Figure 2. ***A, Directions of magnetization of Waterton Formation before magnetic washing. B, Directions of magnetization of Waterton Formation after magnetic washing.***

altered. A second phase of magnetic washing at 250 oersteds did not significantly affect the directions of remanent magnetization, as only changes comparable to the limitations of the astatic magnetometer occurred. The radius of the circles of confidence decreased from 11.5 degrees for the original measurements to 7.0 degrees for the magnetically washed samples, and K increased from 6.5 for the original measurements to 25 for the final measurements, indicating that the dispersion of the vectors had decreased. All samples used were positively polarized (downward directed) except three (DGN: 12S, 14B, 16B), which are negatively polarized (upward directed) and approximately 180 degrees different in declination. For the purpose of statistical analysis these three samples were considered to have their polarity reversed and 180 degrees added to their declination.

Figure 2A is a stereographic representation of the original measurements (Table II). Samples marked 1 and 2 are negatively polarized samples which have been represented as positive poles marked 1 and 2 on Figure 2A. Figure 2B is a stereographic plot of the directions of magnetization for the same samples after magnetic washing. Comparison of these two figures indicates the decrease in the mean declination and a tighter grouping. Sample marked 3 is a negatively polarized sample that was originally positively polarized.

Appekunny Formation

Only two samples of Appekunny strata were collected from Waterton Lakes National Park because of the inadequate exposures of collectable material. The upper part of this formation contains red and maroon quartzites and argillites, but, unfortunately, where outcropped the Appekunny strata were thinly bedded with variations in colour, composition, and texture. The samples collected had a mottled appearance, possibly due to white deoxidation zones indicating reduction of the hematite.

The original direction of the remanent magnetization of the two samples had horizontal components nearly 150 degrees apart. After magnetic washing in a field of 156 oersteds the magnetization direction of one sample was not significantly altered, whereas the remanent vector of the second sample became almost aligned with that of the first. The circle of confidence for this group is 10.0 degrees. Further magnetic washing in a field of 250 oersteds had no significant effect on the magnetization directions of the samples.

Figure 3 is a stereographic plot of the final results shown in Table III.

Table III

Magnetic Measurements for Appekunny Formation

Sample	D	I	Collection Site (2)
DGW 4B	97.5	+62.0	East side of Mount Crandell in Waterton Lakes National Park.
4P	103.5	+58.5	
Mean	100.5	+60.5	Lat. 49°05′, long. 113°55′

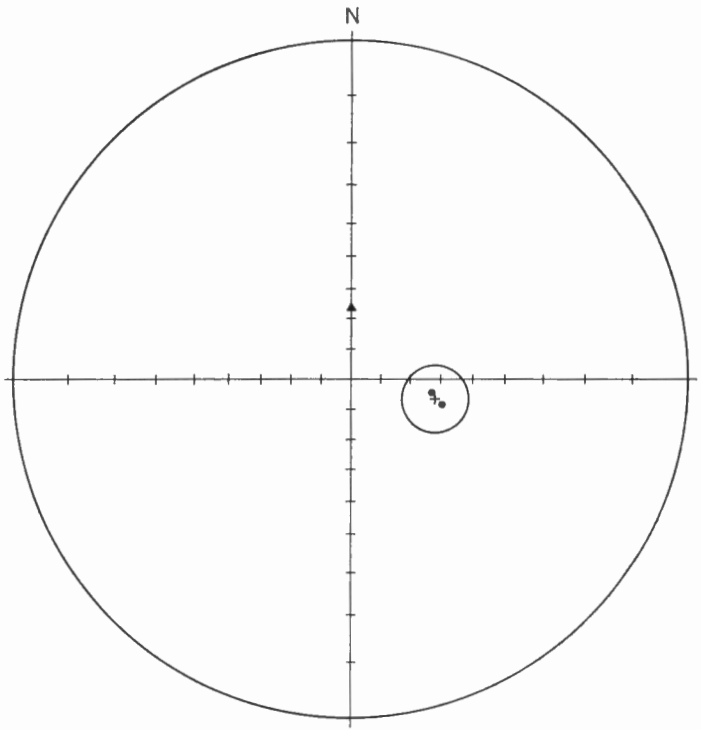


Figure 3. ***Final directions of magnetization of Appekunny Formation.***

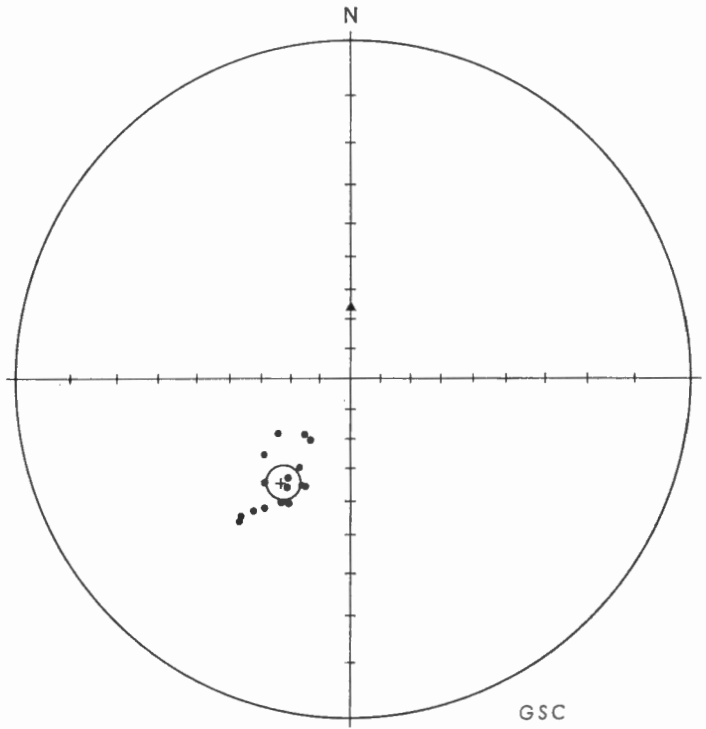


Figure 4. ***Final directions of magnetization of Grinnell Formation.***

Grinnell Formation

The Grinnell Formation, which overlies the Appekunny with a gradational contact, is composed of red argillites, red and white quartzites, and grey sandstones and conglomerates. Its irregular colour banding, thin bedding, and extreme friability made collection very difficult.

Twenty-two oriented samples were collected from three localities (Fig. 1, No. 3). Sixteen of the samples were used in the final analysis, the other six having been rejected for reasons as indicated in Table IV. In general, samples were not affected by magnetic washing at 156 oersteds, but where two cubes from one sample were not in close agreement it was found that magnetic washing brought the two closer together and into better agreement with the majority. All measurable samples, like most of those from the Waterton Formation, were positively polarized.

Figure 4 is a stereographic projection of the final results shown in Table IV.

Table IV

Magnetic Measurements for Grinnell Formation

Sample	D	I	Collection Site (3)
DGP 1B	201.5	+53.0	Northeast part of Pincher Ridge. Lat. 49°17′, long. 114°05′
1S	211.0	+66.5	
2B	202.0	+53.0	
2S	231.5	+60.5	
3B	208.0	+45.5	
3S	213.0	+41.5	
4B	211.5	+53.0	
4S	228.0	+53.5	
5B	218.5	+47.5	
5S	208.5	+57.5	
DGW 1B	209.5	+50.5	Cameron Creek road in Waterton Lakes National Park. Lat. 49°03′, long. 114°01′
1P	205.5	+46.0	
2P	217.0	+34.5	
DGN 7S	217.5	+35.5	Top of Whistler Mtn. on the east side of the Castle River Lat. 49°20′, long. 114°18′
8S	215.5	+39.0	
9S	208.0	+66.5	
Mean	212.5	+50.5	

Rejections

DGW 2B—Very weakly magnetized, and therefore could not be measured accurately.

DGN 7B—This sample was omitted because it is composed of tabular flakes of argillites derived from nearby weathered rocks. (Intraformational breccia.)

DGN 8B, 9B, 10B, 10S—Too weak to be measured with astatic magnetometer, and some were intraformational breccias.

Siyeh Formation

The Siyeh Formation, which lies conformably on the Grinnell Formation, is the thickest in the Purcell sequence where studied. Norris (1959) divided the Siyeh into three parts: the lowest, a grey and green argillite, grey dolomite and limestones as well as grey algal and oolitic limestone; the middle, grey algal dolomite; and the upper, green, grey, and red argillites, molar tooth dolomite, and algal limestone.

Gabbroic Sills in the Siyeh Formation

Two samples were collected from each of two gabbroic sills along a ridge leading to the top of Mount Dungarvan from the south, in Waterton Lakes National Park (Fig. 1, No. 4). The two cubes from each sample generally had a consistent direction of magnetization but from sample to sample a pronounced scatter in both declination and inclination was noted. Magnetic washing had marked effects on two of these samples. The magnetization direction of one sample from each sill remained almost unaltered after a final magnetic washing at 312 oersteds; these two samples remained magnetized in the same steeply plunging direction. The other sample from each sill changed many degrees in declination, and in addition one sample became negatively polarized, the other remaining positively polarized at a low angle. The results indicate that magnetic washing has not completely removed later, superimposed magnetism and the samples are unsuitable, indeed the magnetization direction of two of the samples coincides closely with that of the present earth's field.

The scattered results for such a small number of samples made it impossible to draw any conclusions with respect to the intrusion of the sills into the Siyeh Formation. The figures are, however, given in Table V.

Table V

Magnetic Measurements for Sills in Siyeh Formation

Sample	Original		Final		Collection Site (4)
	D	I	D	I	
DGW 10B	333.5	+ 7.0	97.0	−50.5	Ridge leading from the south to the top of Mount Dungarvan, Waterton Lakes National Park. Lat. 49°07′, long. 113°58′
10P	343.0	+75.5	353.5	+52.5	
11B	320.5	+82.0	359.0	+69.5	
11P	189.0	+12.0	224.0	+ 6.0	

Upper Siyeh

Four samples of red argillite were collected from the top 100 feet of the Siyeh Formation at two separate localities (Fig. 1, No. 5). All samples were magnetized in an almost consistent direction and polarized almost diametrically

opposite to the underlying Grinnell Formation. The direction of remanent magnetization was not significantly altered by magnetic washing in a field of 156 oersteds (Table VI; Fig. 5).

Table VI

Magnetic Measurements for Upper Part of Siyeh Formation

Sample	D	I	Collection Site (5)
DGW 7B	55.5	−28.0	Small exposure above Cameron Creek road, about 50 feet below contact with Purcell Lava (Waterton Lakes National Park). Lat. 49°04′, long. 114°01′
7P	49.5	−27.0	
DGP 9B	47.0	−23.5	Small exposure about 25 feet below contact with Purcell Lava on northwest arm of Pincher Ridge. Lat. 49°17′, long. 114°07′
9S	32.5	−17.5	
Mean	47.0	−24.0	

Purcell Lava

The Purcell Lava, a dark purplish green basaltic flow approximately 300 feet thick, lies directly above the Siyeh Formation. This was the most widely collected formation—twenty-eight samples collected from five different localities—as resistant exposures were accessible and well jointed. The geological tilt of the sample in situ was estimated from the average strike and dip of the underlying sedimentary beds. The remanent magnetic moment reported for the formation accordingly refers to the palaeohorizontal plane at the time of formation of the lava flow. Most of the results quoted in Table VII are for samples that have been progressively washed magnetically to such a degree that further washing had no significant effect on the direction of remanent magnetization. In other samples significant changes continued to occur to the limits of the apparatus; these were not used. The original direction of magnetization for many samples was greatly altered by the progressive washing.

Figure 6 is a stereographic plot of the results shown in Table VII, with the exception of the samples collected from the northwest arm of Pincher Ridge, indicated as DGP. Rejection of these samples was felt to be justified when each group of samples was analyzed and compared. Those marked as DGN (1-6), DN (1-3), DN (7-9) and DGW (5-6) were found to be magnetized in a consistent direction, one with another, whereas the directions of those marked as DGP varied widely. Figure 6 illustrates the scatter of the results obtained for the selected samples and the large circle of confidence, 18.5 degrees, calculated from an analysis of the results.

This scatter may be attributed to hydrothermal alteration of the minerals, as chlorite-magnetite-leucoxene pseudomorphs have been observed which suggest that

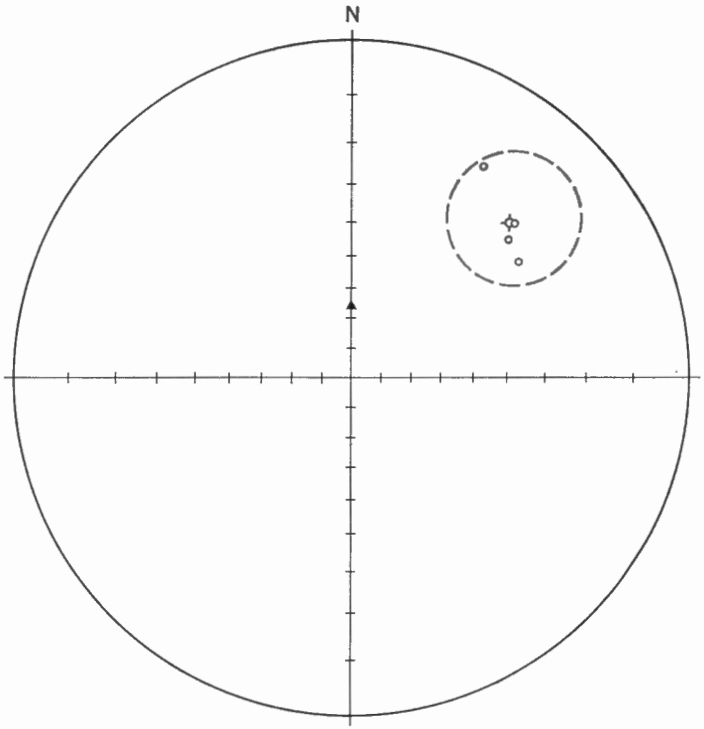


Figure 5. *Directions of magnetization of Siyeh Formation (upper part).*

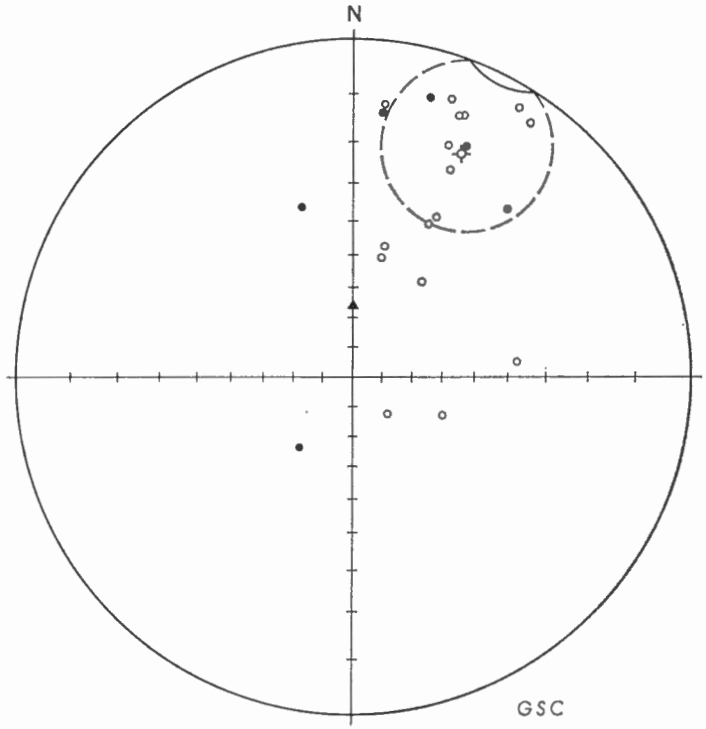


Figure 6. *Directions of magnetization of selected samples of Purcell Lava. Corrected for geological tilt.*

a titaniferous clinopyroxene has been replaced. The pseudomorphic relationships and the distribution of the magnetite suggest that it is partly secondary after ilmenite, that this chemical alteration, a result of physicochemical change, possibly took place during cooling of the magma as is suggested by the approximate coincidence of the direction of magnetization with that of the sedimentary formations. Another cause of the scatter may be that more than one lava flow was involved, so that one flow may have affected another as it cooled below its Curie point.

Table VII

Magnetic Measurements for Purcell Lava

Sample	D	I	Collection Site (6)
DGP 6B	52.0	+ 4.0	Small exposure on northwest arm of Pincher Ridge.
6S	55.0	− 4.5	Lat. 49°17′, long. 114°07′
7B	205.0	+10.5	
7S	62.5	+ 3.0	
8B	12.0	+33.0	
8S	204.5	− 8.5	
Mean	70.0	+17.5	
DGN 1B	6.5	−12.0	Ridge of exposed lava on east side of Castle Rive , between
1S	26.0	+16.0	Whistler Mtn. and Mount Gladstone from elevations
2B	343.0	+34.5	5,200 feet to 6,500 feet.
2S	26.5	−36.0	Lat. 49°19′, long. 114°17′30″
3B	15.0	+ 9.0	
3S	215.0	+62.0	
4B	111.0	−58.0	
4S	13.0	−50.0	
5B	13.0	−47.0	
5S	42.0	+22.0	
6B	132.0	−73.0	
6S	34.5	−51.0	
Mean	13.0	−23.0	
DN 1	22.5	−10.0	Collected by D. K. Norris from an exposure of lava near
2	24.5	−22.0	North Kootenay Pass at headwaters of Carbondale
3	27.0	−33.5	River.
			Lat. 49°23′30″, long. 114°35′
Mean	24.5	−22.0	
DN 7	22.5	−10.0	Collected by D. K. Norris from lava exposed on Table
8	19.0	− 8.0	Mountain east of Castle River.
9	31.0	− 4.0	Lat. 49°20′, long. 114°18′
Mean	24.0	− 7.5	

Table VII—(Cont'd.)

Sample	D	I	Collection Site (6)
DGW 5B	84.0	−37.5	From an exposure above Cameron Creek road, Waterton Lakes National Park. Lat. 49°04′, long. 114°01′
5P	6.0	+14.5	
6B	34.5	− 5.0	
6P	22.5	−17.0	
Mean	32.0	−13.0	

Kintla Formation

Directly overlying the Purcell Lava is the Sheppard Formation, a brown weathering, pale greyish green dolomite, siltstone, and sandstone, a few hundred feet thick, which is in turn overlain by the Kintla Formation divided by Hage (1943) into four members, A, B, C, and D. Fine-grained red sandstones and siltstones were collected from Members A and C, and four samples from a gabbroic sill intruded into Member B.

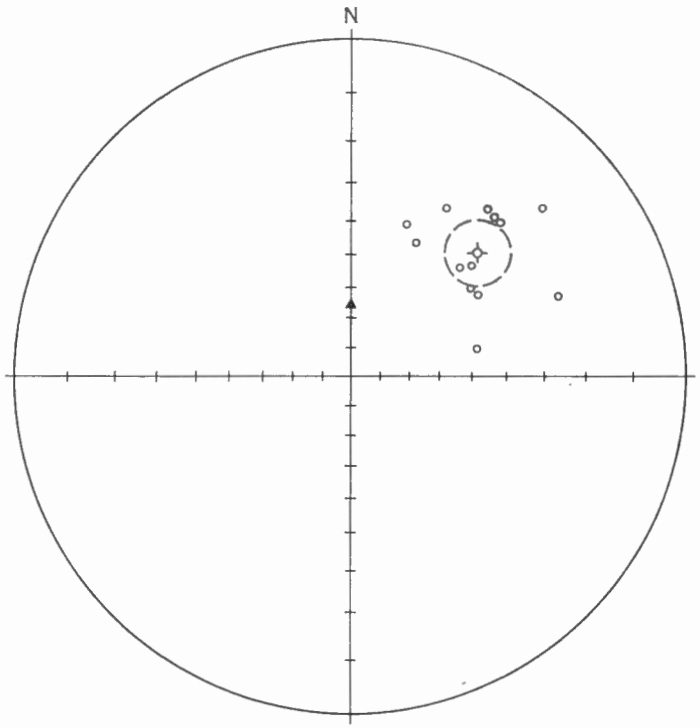
Kintla Formation, Member A

Thirteen oriented samples of red sandstone and siltstone were collected from three localities in this member. These samples were not significantly altered by magnetic washing in a field of 156 oersteds. Figure 7 is a stereographic plot of the direction of magnetization for the samples given in Table VIII. All the samples, it may be noted, are reversely polarized and in approximate agreement with the results obtained for the upper Siyeh and the Purcell Lava.

Table VIII

Magnetic Measurements for Kintla Formation, Member A

Sample	D	I	Collection Site (7)
DGW 12B	48.0	−37.5	Exposure in ditch approximately 6 miles west of Waterton Park village along Cameron Creek Road. Lat. 49°03′, long. 114°01′
12P	20.5	−38.5	
13B	27.0	−42.5	
13P	78.0	−48.0	
14B	44.0	−25.0	
14P	45.0	−40.0	
15B	69.5	−23.5	
15P	57.5	−41.5	
DN 4	43.0	−25.0	Collected by D. K. Norris near North Kootenay Pass at headwaters of Carbondale River. Lat. 49°23′30″, long. 114°35′
5	49.0	−15.5	
6	40.0	−24.5	
DN 10	30.0	−30.0	Collected by D. K. Norris from exposures on Table Mountain east of Castle River. Lat. 49°20′, long. 114°18′
11	53.5	−42.0	
Mean	46.0	−34.0	



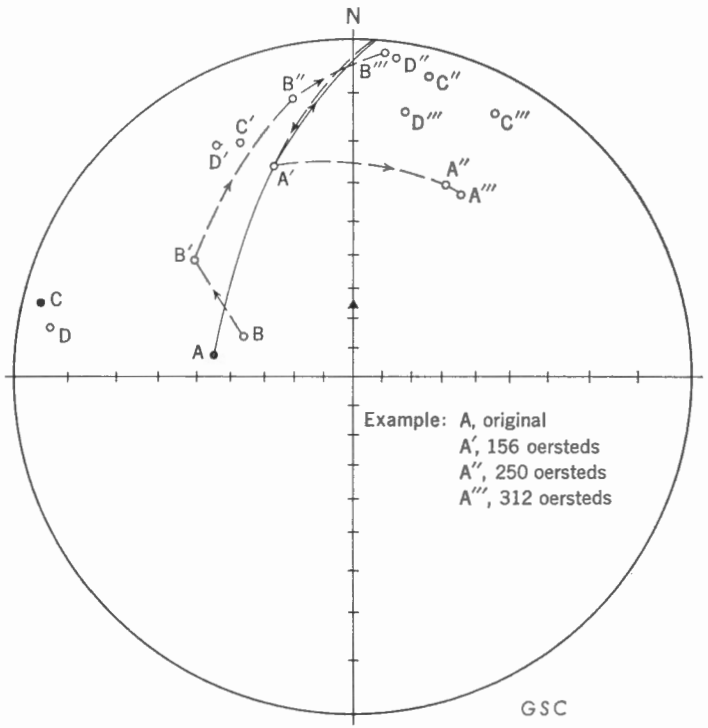
LEGEND

North pole on upper hemisphere . . . . . o

Mean direction with circle of confidence . . . ✧

Present theoretical dipole . . . . . ▲

Figure 7. *Directions of magnetization of Kintla Formation, Member A.*



LEGEND

North pole on lower hemisphere . . . . . ●

North pole on upper hemisphere . . . . . o

Present theoretical dipole . . . . . ▲

Figure 8. *Effects of magnetic washing on samples from sill in Kintla Formation, Member B.*

Sill in Kintla Formation, Member B

Four samples were collected from a medium- to fine-grained gabbroic sill lying conformably in Member B about 300 feet below its contact with Member C. The original magnetization directions of these samples were greatly scattered, two of the samples being normally polarized and the other two reversely polarized. Figure 8 is a stereographic projection of the magnetization vectors after different degrees of magnetic washing. The great circle traced by two of these vectors in moving from their original position to another is also shown. It is of interest to note that the changes brought about by magnetic washing, though large, are consistent in that all poles moved in the same direction and all samples became negatively polarized after being exposed to an initial demagnetizing field of 156 oersteds. It is also evident that only one sample (A) appears to have reached a final position (A′′′) as indicated by the insignificant change from the position occupied after washing at 250 oersteds to that occupied after a further magnetic washing in a field of 312 oersteds. This is not true of the other three samples, as there was a significant movement produced by the final washing. The maximum demagnetizing field that could be applied with the available apparatus was 312 oersteds.

These results show that (1) the magnetization directions coincide approximately with those of the reversely polarized rocks of the underlying formations; (2) the rocks may have been intruded while the polarity of the earth's field was reversed, and before the deposition of the C member which, as shown later, is positively polarized; or (3) they may have been intruded after the deposition of Member C when the earth's field may again have been reversed. It must be pointed out that as the sedimentary rocks of Member B have not themselves been measured, the negative polarity may be due to a physicochemical reaction within the sill itself (Néel, 1955).

Table IX

Original and Final Magnetic Measurements for Sill in Member B

Sample	Code on Fig. 8	Original		Code on Fig. 8	Final		Collection Site (8)
		D	I		D	I	
DGN 19B	A	278.5	+45.5	A′′′	30.5	−26.0	All samples collected from a single exposure of sill on ridge running from north-end of Windsor Mountain to Mount Gladstone.
19S	B	289.5	−52.5	B′′′	5.5	− 2.5	
20B	C	283.0	+ 3.5	C′′′	28.5	− 7.5	
20S	D	279.0	− 6.0	D′′′	11.0	−13.0	
Mean					18.0	−12.5	Lat. 49°18′40″, long. 114°14′

Kintla Formation, Member C

Twenty-five samples of highly quartzose red sandstones and siltstones were collected from Member C at two localities. All the samples were magnetically washed in a demagnetizing field of 156 oersteds and most were unaffected. Samples for which the magnetization direction of the two cubes originally differed significantly after magnetic washing agreed with each other and with those on which magnetic washing had little effect. From Table X and Figure 9 it will be seen that all the samples are positively polarized and have a consistent magnetic declination which is comparable to that of the underlying, positively polarized, formations. The samples DGN 21 to DGN 31 represent a collection made across the total exposed thickness of Member C.

Table X

Magnetic Measurements for Kintla Formation, Member C

Sample	D	I	Collection Site (9)
DGN 21B	219.5	+62.0	Collected along a 1-mile exposure on ridge leading from east bank of Castle River to north end of Windsor Mountain Long. 114°14′ 114°15′30″ Lat. 49°18′20″
21S	221.0	+42.0	
22B	214.5	+59.0	
22S	220.5	+57.0	
23B	245.0	+62.0	
23S	232.5	+64.0	
24B	216.0	+54.5	
24S	209.0	+58.0	
25B	219.0	+39.0	
25S	219.0	+37.5	
26B	196.5	+51.0	
26S	213.0	+46.0	
27B	210.0	+43.5	
27S	215.0	+56.0	
28B	213.0	+28.0	
28S	226.0	+ 9.5	
29B	217.5	+58.0	
29S	227.5	+51.5	
30B	283.0	+85.0	
30S	227.0	+60.0	
31B	207.0	+27.5	
31S	212.0	+39.0	
PFA 161	278.0	+21.0	Collected by R. A. Price from Galton Range, Fernie map-area, 40 miles west of Waterton Lakes National Park. 161– Long. 114°48′41″, lat. 49°1′39″ 219– Long. 114°58′58″, lat. 49°2′04″ 263– Long. 114°58′45″, lat. 49°10′16″
219	250.0	+33.5	
263	169.5	+40.0	
Mean	220.5	+49.0	

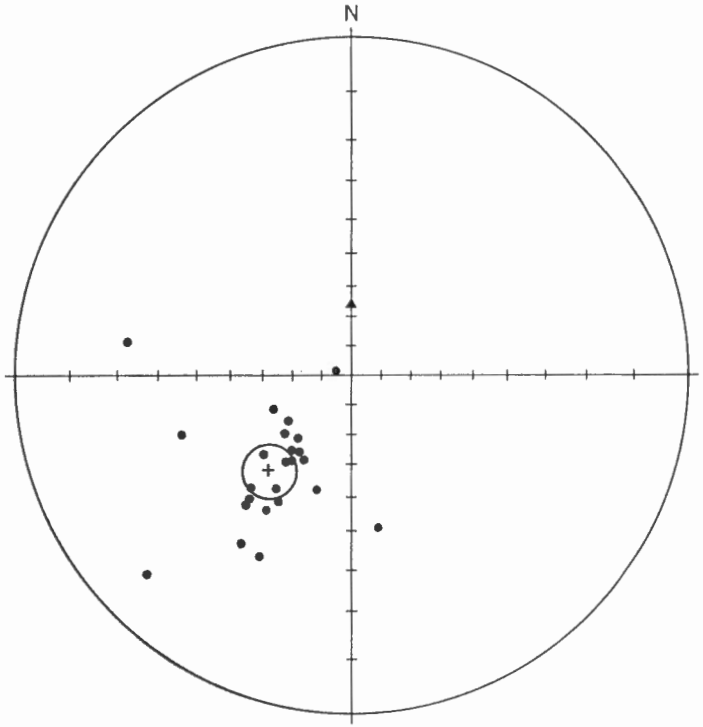


Figure 9. *Directions of magnetization of Kintla Formation, Member C.*

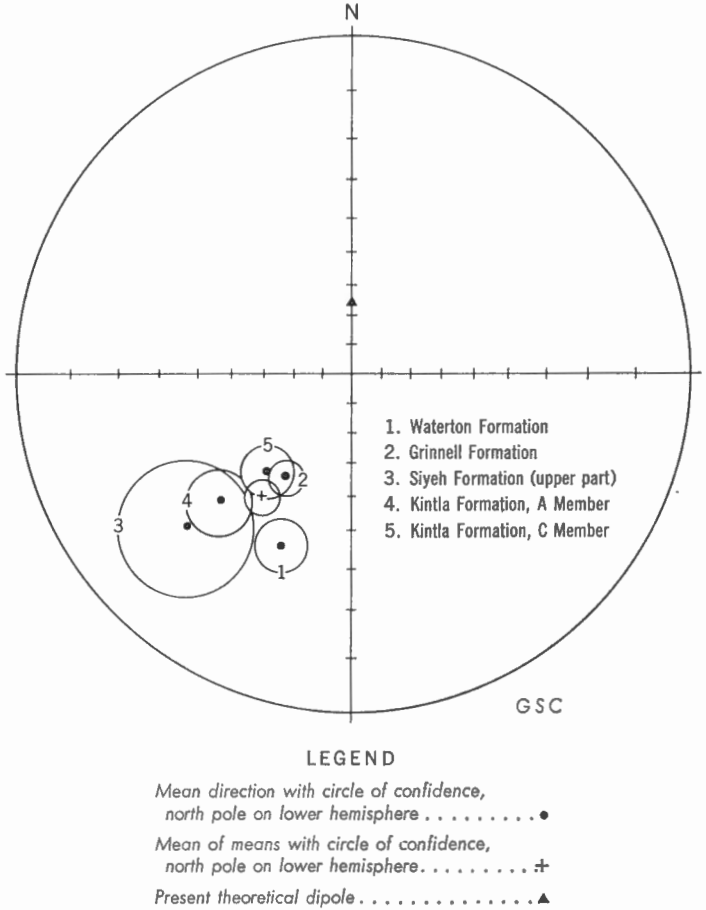


Figure 10. *Mean directions of magnetization for stable formations, and mean of means.*

Summary of Results

Table XI shows the mean directions of magnetization, the coordinates of the ancient pole positions, and the statistical parameters Θ, K, and N, compiled together for comparison, for all formations measured.

Table XI

Summary of Palaeomagnetic Data for each of the Formations Studied

Formation or Member	Collection Site	D	I	Pole Position		Θ	K	N
				Long.	Lat.			
Kintla C	9	220.5	+49.0	148.5W	3.0S	8.0	14	25
Sill in Kintla B	8	018.0	−12.5	136.0W	31.0S	19.0	24	4
Kintla A	7	046.0	−34.0	158.5W	11.5S	8.5	25	13
Purcell Lava	6	025.5	−17.5	144.0W	28.5S	18.5	4	22
Upper Siyeh	5	047.0	−24.0	162.0W	16.0S	16.0	34	4
Sills in Siyeh	4							
Grinnell	3	212.5	+50.5	142.5W	5.0S	5.0	52	16
Appekunny	2	100.5	+60.5	120.0E	24.0S	10.0	645	2
Waterton	1	202.0	+33.0	137.5W	20.0S	7.0	25	17

DISCUSSION OF RESULTS

The first conclusion that may be reached from a study of these results is that the direction of the earth's magnetic field appears to have been reversed during the formation of the upper Siyeh, Purcell Lava, Kintla A, and possibly Kintla B. The fact that both sedimentary and igneous rocks, which have totally different compositions, are polarized in an upward direction strongly suggests that the reversal in these formations was due to the reversal of the earth's field rather than to a self-reversal mechanism.

Again, from an analysis of the pole positions for the stable formations in the Purcell System, it is evident that the position of the pole wandered westward from its position in the Waterton up to its position in the Kintla A and then shifted slightly eastward.

It is also evident from a statistical analysis of the palaeomagnetic results that the magnetization of some formations is stable, with a common direction, whereas that of others is unstable. The Waterton, Grinnell, upper Siyeh, and Kintla A and C, for instance are considered to be stably magnetized as the direction of magnetization in them is almost consistent and magnetic washing did nothing but improve the consistency of the data. Samples and/or specimens that originally differed from the normal grouping were brought into line by the magnetic washing, and those that originally fell into this grouping were not significantly affected by it. However, in samples from the Grinnell Formation that contained inclusions of tabular argillite plates derived from nearby red beds, the direction of magnetization was quite different from that of the overall group. This is to be expected because, although the fine-grained matrix would have a magnetic vector controlled by the earth's field at the time of deposition, the magnetization direction of the chips would be that of the formation from which they were derived. Nor would the magnetization direction of the chips be entirely at random, as the chips themselves tend to be oriented parallel with the bedding. The combining of the two vectors, that of the matrix and that of the chips, would produce an anomalous resultant. If the formation had been completely remagnetized after formation there would, of course, not be this discrepancy between the fine-grained sediments and the breccia.

The results for the Appekunny Formation are not acceptable because (1) only two samples were secured, (2) there was visible alteration on the surface of the specimens, and (3) the direction of magnetization differs greatly from that of the overlying and underlying beds of the Grinnell and the Waterton Formations, from which an adequate number of samples were obtained. Similarly the results for the two fine-grained gabbroic sills in the Siyeh Formation were rejected as unsuitable. The results obtained from the Purcell Lava are not encouraging, as seen from the scatter of the directions of magnetization and the large circle of confidence, 18.5 degrees. However, it is interesting to note that the Purcell Lava and the overlying and underlying beds are reversely polarized and have approximately the same mean direction of magnetization. The results from

the sill in Kintla B are palaeomagnetically unsuitable as it is not certain that the final direction of magnetization for three of the four samples was reached. This is indicated by the changes in direction of magnetization between the second last and final magnetic washing. Nonetheless the results do coincide approximately with those for the Purcell Lava.

Figure 10 is a stereographic plot of the mean directions of magnetization and shows the circles of confidence, calculated at the 95 per cent level, for the formations given in Table XI except for the Appekunny, the Purcell Lava, and the sill in B member of the Kintla Formation. The overlapping or close proximity of the circles of confidence for the stable formations indicates that there is little if any significant difference in mean directions of magnetization for these formations. It therefore may be concluded that movement of the geomagnetic pole during the deposition of that part of the Purcell System studied, which is approximately 8,000 feet thick, has been within the limits set by the scatter of the mean directions of magnetization for the various stable formations and is therefore insignificant. For this reason a mean direction of magnetization is calculated from seventy-five samples from the Waterton, Grinnell, and upper Siyeh Formations, and Members A and C of the Kintla Formation, representing the total exposed thickness within the collecting area, and this is used to calculate the position of the ancient geomagnetic pole relative to North America for the Purcell System (Table XII). The interpretation given is therefore based upon a statistical analysis for the Purcell System as a whole rather than on individual formations. Unless a systematic set of samples were collected across a sufficient thickness of strata, errors might be introduced due to short period variations in the earth's magnetic field.

Table XII

Mean Palaeomagnetic Results for the Purcell System

(*see* Fig. 11 and Table XIV)

Formations	D	I	Pole Position		Θ	K	N	$d\psi$	$d\chi$
			Long.	Lat.					
Kintla C..... Kintla A..... Upper Siyeh..... Grinnell..... Waterton.....	216.0	+42.5	148.0W	10.0S	5.5	10	75	4.2	6.8

Collection site 114°W, 49°N.

Measurements from the Belt Series obtained by Collinson and Runcorn (1960) from samples collected 2°S of the writer's collection area, have given results that reasonably agree with those obtained by the author (Table XIII and Fig. 11).

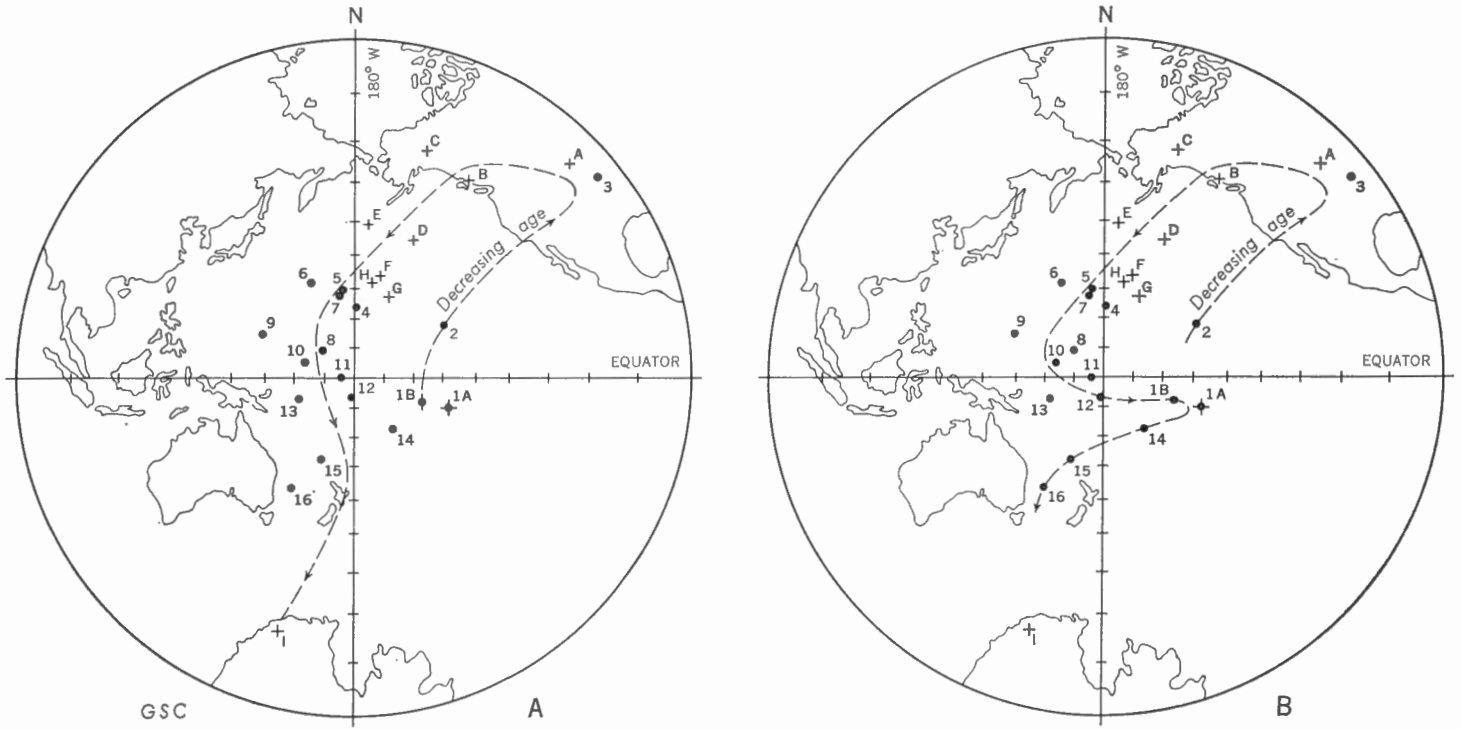


Figure 11. ***A, Interpretation of polar wandering curve, Path 1***
B, Interpretation of polar wandering curve, Path 2

Table XIII

Palaeomagnetic Data for Formations of the Belt Series

(Collinson and Runcorn, 1960)

Formation	D	I	θ	N	Pole Position		Collection Site	
					Long.	Lat.	Long.	Lat.
McNamara	026	−43	4	20	138W	15S	114W	47N
Miller Peak	234	+30	7	14	168W	11S	—	—
Spokane	220	+52	4	39	147W	2S	—	—
Grinnell	225	+48	6	16	153W	4S	—	—
Appekunny	223	+29	6	15	158W	17S	—	—
Mean	227	+41	—	—	157W	8S		

Spokane Formation

According to Fenton and Fenton (1937), the Spokane Formation consists of argillaceous and arenaceous strata interbedded with green basalts, and lies above the Siyeh Formation and below the Sheppard Formation.

Miller Peak Formation

A thinly bedded, dominantly green and red formation, which represents the initial stage of Missoula clastic sedimentation. It is correlated by Fenton and Fenton (1937) with the Kintla Formation, which is referred to as a member of the Miller Peak Formation.

McNamara Formation

A 3,000-foot formation overlying the Hellgate Formation, which in turn overlies the Miller Peak Formation (Clapp and Deiss, 1931).

Age and Correlation of the Purcell (Belt) System

The pole position computed for the Purcell System does not coincide with any previously calculated for Precambrian times except those reported by Collinson and Runcorn (1960) for the Belt Series 2°S of the writer's area. Correlation of these pole positions with those of other North American Precambrian formations, specifically those of Keweenawan age (DuBois, 1962), has, in the writer's opinion, two possible interpretations. Pole positions for sixteen late Precambrian formations listed in Table XIV are plotted on Figures 11A and B (full circles and numbers). These are for rocks of Keweenawan age as interpreted by DuBois (op. cit.) and are those directly concerned with correlation of the Purcell System. For the sake of completeness, nine other poles are plotted (crosses and letters). The dashed line in Figure 11A is the complete generalized polar wandering curve as interpreted by DuBois (op. cit.), that on Figure 11B is an alternative path suggested by the writer. Poles 1A and 1B are approximately 25°S of the pole position computed for the Sibley Series (pole 2) by DuBois.

The Duluth Gabbro (pole 4) of middle Keweenawan age has been dated (Goldich, *et al.*, 1957) at 1,120 million years, which according to the interpretation of Figure 11A is younger than the Purcell System (Belt Series). Age determinations on rocks of the Belt Series have been made but do not conclusively date the time of Beltian deposition. An age determination by Eckelmann and Miller (1956) made on a pitchblende vein cutting the St. Regis Quartzite (Belt Series) near Coeur d'Alene, Idaho, has dated the pitchblende as being at least 1,200 million years old. This agrees with isotopic age determinations by Wanless and Leech (1958) obtained from twenty-one samples taken from the Sullivan ore deposit at Kimberley, British Columbia, which is contained in a host rock of Purcell age. The average common lead age was calculated as 1,255±100 million years. All these suggest a minimum age for the Purcell System of not less than 1,200 million years.

Table XIV

Late Precambrian Pole Positions

No. on Figure 11	Formation, System, or Series	Pole Position		Determined by
		Long.	Lat.	
16	Umfraville and Thanet	157.8E	34.2S	Hood
15	Lodore	169.0E	27.0S	Collinson and Runcorn
14	Jacobsville (Keweenaw)	166.0W	17.5S	DuBois
13	Grenville	162.0E	7.0S	DuBois
12	Mean Jacobsville	179.0E	6.5S	DuBois
11	Orienta I.	176.0E	0.0	DuBois
10	Jacobsville (Sault)	164.0E	5.0N	DuBois
9	Eileen	150.5E	13.0N	DuBois
8	Freda and Nonesuch	170.0E	9.0N	DuBois
7	Hakatai	175.0E	27.0N	Collinson and Runcorn
6	Sault Freda (?)	164.5E	31.0N	DuBois
5	Copper Harbor	176.5E	29.0N	DuBois
4	Duluth Gabbro	179.5W	23.5N	DuBois
3	Alona Bay Lavas	95.0W	39.0N	DuBois
2	Sibley Series	149.0W	16.0N	DuBois
1B	Belt Series	157.0W	8.0S	Collinson and Runcorn
1A	Purcell System	148.0W	10.0S	Black
Letter				
I	Chequamegon	131.5E	68.0S	DuBois
H	Mamainse Point Lavas	173.5W	31.5N	DuBois
G	Portage Lake	168.5W	26.5N	DuBois
F	Logan diabase II	171.5W	33.0N	DuBois
E	Hazel	175.0W	49.0N	Martinez, Howell, Statham
D	Boulter gabbro	156.8W	42.6N	Hood
C	Levack	140.7W	64.0N	Hood
B	Logan diabase I	130.0W	53.5N	DuBois
A	Baraga Co. dykes	99.5W	45.0N	Graham

The alternative polar wandering curve, path 2, as suggested by the writer and shown in Figure 11B, has been extended eastward after passing pole 4 to include the pole position for the Purcell System. If this interpretation is correct, the Purcell System is about the same age as the Jacobsville from the Keweenaw Peninsula (pole 14), which is considered to be late Keweenawan, about 1,000 million years—roughly the age of the metamorphism of the rocks of the Grenville subprovince (pole 13). Age determinations also support this path. Kerr and Kulp (1952) report ages for the uraninite, also from the Sunshine Mine at Coeur d'Alene district, Kellogg, Idaho, as being 710 ∓ 10 million years for Pb^{206}/U^{238}; 750 ∓ 10 million years for Pb^{207}/U^{238}; and 850 ∓ 50 million years for Pb^{207}/Pb^{206}. If it is assumed that the major source of divergence in these ages is attributed to radon loss, then a more reliable age is 750 ± 50 million years. Age determinations by Goldich, *et al.* (1959) on an illite-bearing shale in the Siyeh Formation, collected at Logan Pass, Glacier Park, Montana, have given ages for the formation of the illite as being 740 million years by the potassium-argon method and 780 million years by the strontium-rubidium method. These datings, of course, represent ages that are definitely younger than Beltian, for they give the time of formation of the uraninite-bearing veins or of the metamorphism that produced the illite.

Neither the pole positions nor the age determinations conclusively date the time of formation of the Purcell System (Belt Series) but they do limit two possibilities. If path 1 (Fig. 11A) is correct, the Purcell System (Belt Series) is definitely older than the middle Keweenawan Duluth Gabbro (1,120 million years), possibly a good deal older than 1,200 million years, and older than the Sibley Series of early Keweenawan age. If path 2 (Fig. 11B) is correct, then Purcell (Beltian) sedimentation took place during late Keweenawan time 900-1,000 million years—about the time the Jacobsville Formation of the Lake Superior region was deposited—and is therefore definitely younger than the Duluth Gabbro.

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